ALS Disease Progression

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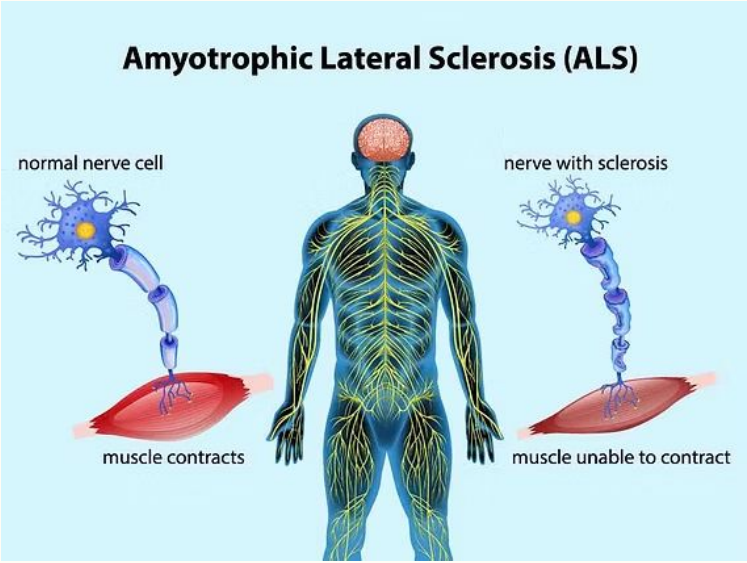
DATS 4001 Capstone Demo

Central Problem

- What is ALS?
- ALS progression differs across patients
- Disease trajectory is difficult to predict from a single snapshot
- Help researchers find new methods to estimate progression over time



Data Sources

PRO-ACT: primary modeling dataset

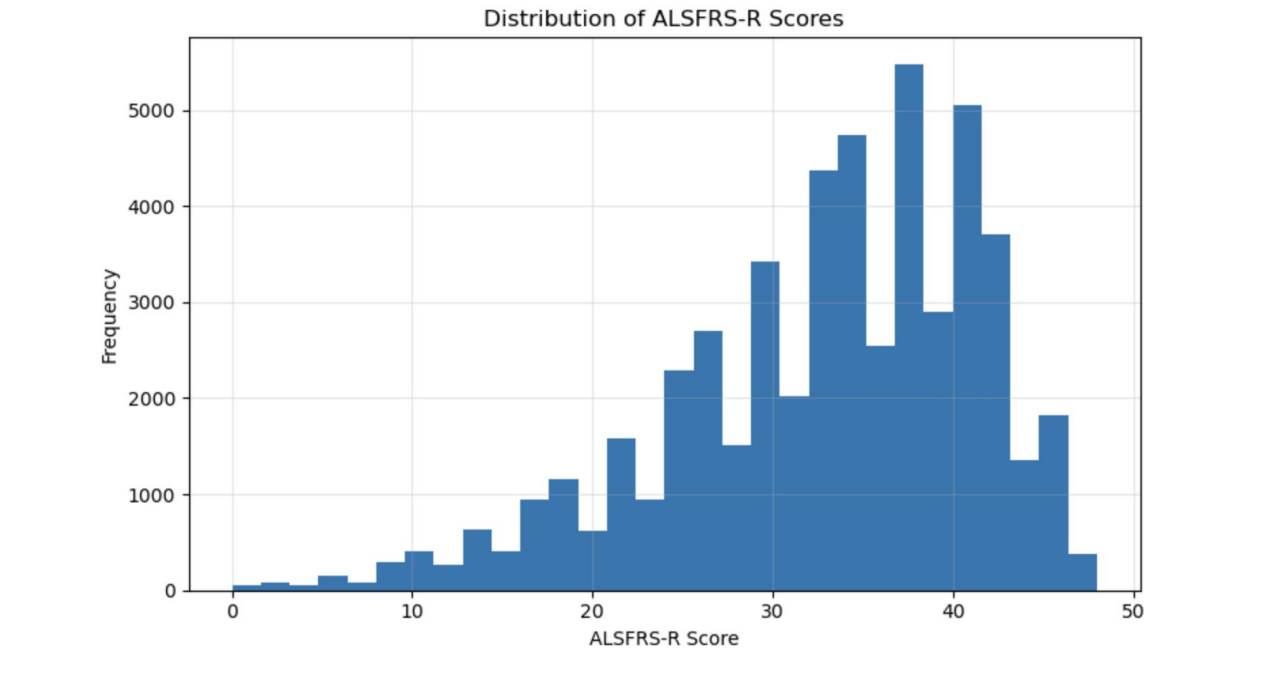
Target ALS: supplementary clinical data

*de-identified longitudinal patient records

Outcome Variable: ALSFRS-R

Clinical functional scale for ALS severity

Score range: 0 to 48



Workflow Overview

1. Data processing
2. Exploratory analysis
3. Feature engineering + modeling
4. Evaluation + interpretation

Data Processing Steps

1. Loaded structured patient visit data
2. Standardized column names
3. Selected core modeling variables
4. Created longitudinal visit structure

Outcome Variable: ALSFRS-R

alsfrs_delta

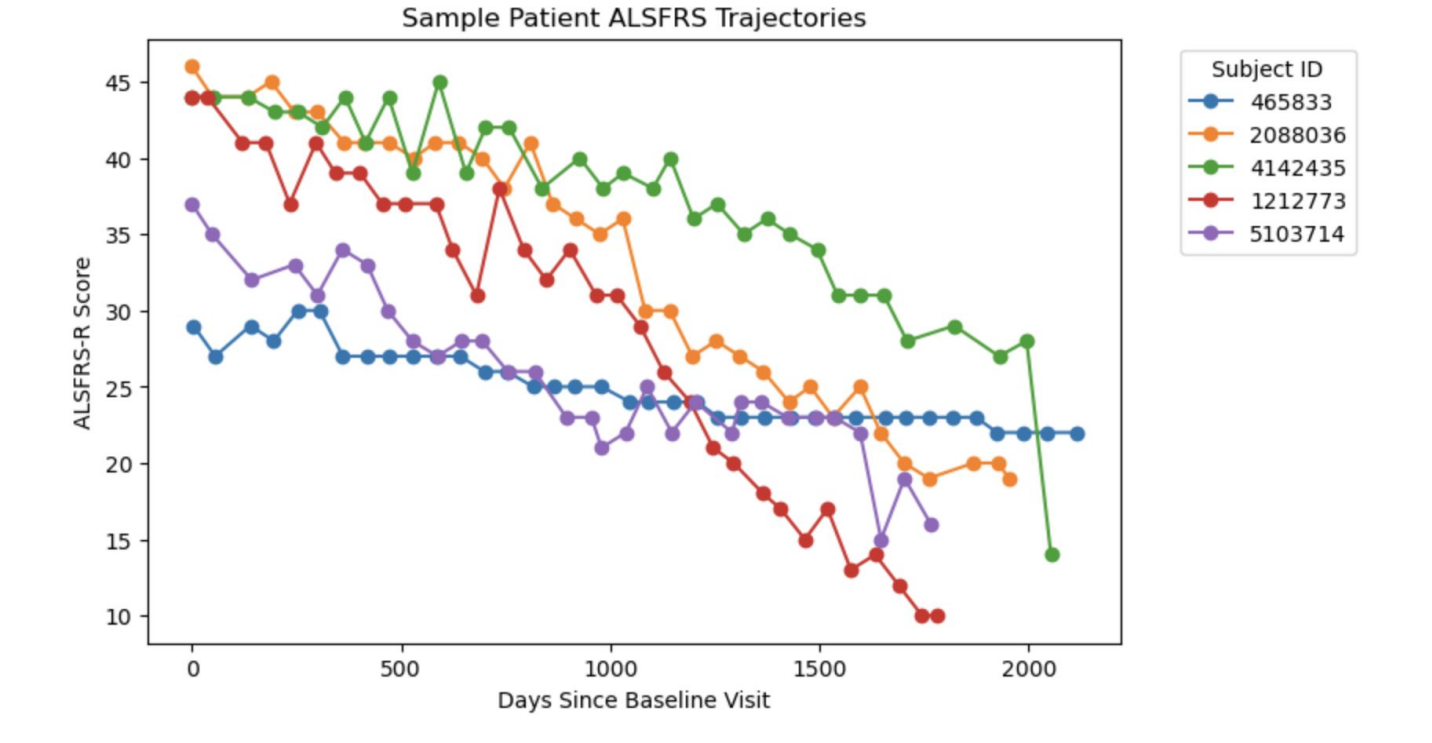
prev_alsfrs_r_total

baseline_alsfrs_r_total

delta_from_prev_visit

time_since_first_visit

EDA: Patient Trajectories



Modeling Setup

Models

- Naive baseline
- Linear regression
- Random forest
- Gradient boosting

Evaluation Metrics

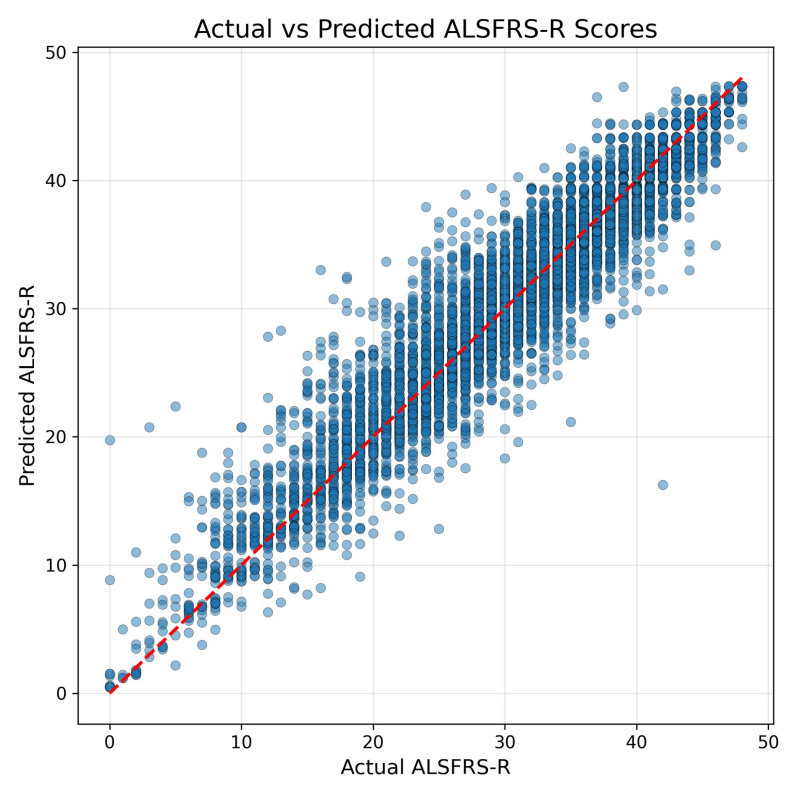
- MAE
- RMSE
- R^2

Model Comparison

	model_name	mae	rmse	r2
0	naive_previous_visit	1.939792	3.012288	0.882153
1	linear_regression	1.829816	2.618816	0.910929
2	random_forest	1.821809	2.599467	0.912241
3	gradient_boosting	1.805794	2.573580	0.913980

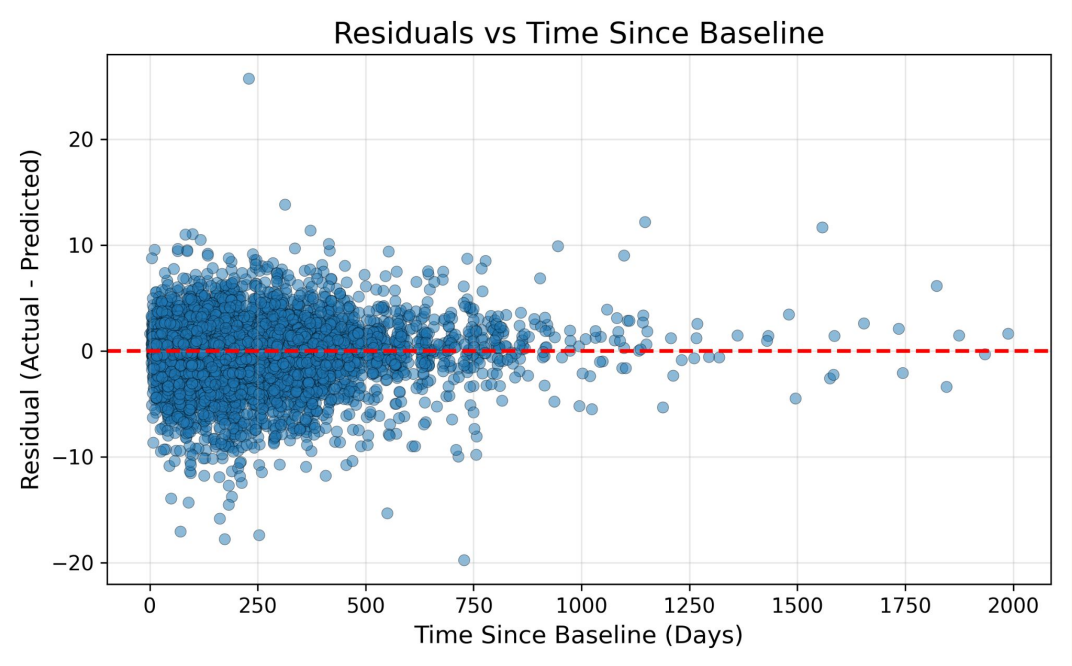
Visuals

Actual vs Predicted (RF)

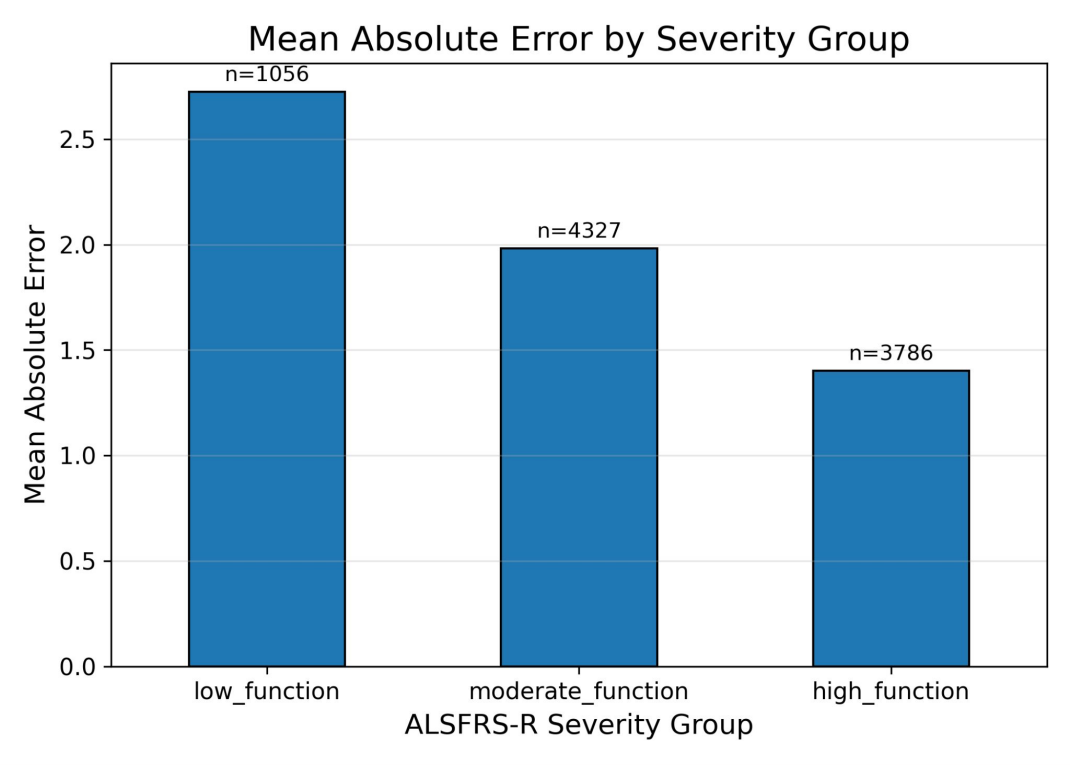


- $R^2 \approx 0.91$
- MAE ≈ 1.83

Residuals vs Time



Error by Severity Group



Results

Main Findings

- Longitudinal patient history is highly predictive
- Machine learning models outperform the naive baseline
- Model accuracy drops for lower-function patients

Limitations

- Missing clinical information
- Limited feature richness
- Strong dependence on observed prior visits

Thank you!